Uniqueness for a Pell equation

Problèmes ouverts — A. Dujella · 2026

Problem 4.1 (Dujella, 2026). Let $k \geqslant 2$ be an integer. Is it true that the Pellian equation

$$x^2 - (k^2+1)y^2 = k^2$$

has at most one solution with $0 < y < k-1$ and $x > 0$?

Remark. Matthews, Robertson and White (Glas. Mat. Ser. III **48** (2013), 265–289) verified that the statement is true for $k \leqslant 2^{50}$, while Le and Srinivasan (Glas. Mat. Ser. III **58** (2023), 59–65) proved it for all k of the form $k = p^m q^n$, where p, q are distinct odd primes and m, n are positive integers.